# MechMonics (Mechanical harMonics)

## Discrete Harmonics for Polycrystals

**Purpose:** To provide grain-based harmonic modes for use in reduced-order representations of field data.

**Approach:**

- For a virtual polycrystal discretized with finite elements, solve for a set of discrete harmonic modes for every crystal of the virtual polycrystal.
- Determine grain axes for each grain.
- Evaluate a set of weights corresponding to the set of modes to approximate existing field data over each crystal.
- Assess the error of the expansion representation based on the difference between the original field data and the expansion.

**Methodology:**

- Using a MatLab-based finite element code, determine grain-based discrete harmonic modes by solving for singular values of the negative Laplacian.
- Import virtual polycrystals instantiated and meshed using **Neper**.
- Supplement with **MechMet** to obtain mechanical metrics of the polycrystal.
- Export results to VTK files for visualization with **Paraview** or **VisIt**.

**Reference and Citation:**

**P. R. Dawson and M. P. Miller. Mechanical analyses using discrete harmonic expansions of grain-based field data. arXiv:2103.05483v1 [cond-mat.mtrl-sci], 2021.**

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## Methodology

Representation of a field using a finite element representation takes the form:

$$a(\mathbf{x}) = [N(\mathbf{x})]\{A\}$$

where $[N(\mathbf{x})]$ are the interpolation functions and $\{A\}$ are the nodal point values.

The nodal point values are approximated with an expansion:

$$\{A\} = \sum_k \beta^k \{U^k\}$$

where the basis functions are determined from solutions of the negative Laplacian:

$$-\nabla^2 u = \lambda u$$

Modes are the singular values and vectors: $(\lambda^i, \{U^i\})$ (MatLab's SVD routines are invoked for this purpose.)

Defining features:

- Finite element discretization using **Neper** tessellation of a polycrystal
- Computations performed grain-by-grain using a modification of **Neper** mesh
- Matrix equation depends only on grain geometry (no physical constants)
- Users specify the number of modes desired
- Uses field data from simulation or experiment projected onto nodes of the mesh
- Users may track the expansion of weights ($\beta^k$) using time history of field data

# MechMonics (Mechanical harMonics)

## How to Run MechMonics

**Installation:** download source files;
unpack in a separate directory;
set MatLab path to include directory.

**Executable scripts:**

- MechMonics.m
- FitFieldDatawithHarmonics.m
- PlotBetaEvolution.m

*Input and output parameters are defined in the scripts.*

**Required input data:**

- Neper mesh file (including grain definition and grain orientations)
- Field data at every nodal point of the mesh (only when using FitFieldDatawithHarmonics)

### MechMonics:

Execute with: MechMonics()

Run this script first to compute the harmonic modes and grain axes for each grain.

### Analysis of simulated field data* using MechMonics:

#### FitFieldDataWithHarmonics:

Execute with: [beta_all_orth,error_all_orth,beta_all_opt,error_all_opt,grain_volumes] = FitFieldDataWithHarmonics()

Run this script after MechMonics to compute the harmonic mode weights for each grain for stress fields.

#### PlotBetaEvolution:

Execute with: PlotBetaEvolution(beta_all,grlist)

Run this script after computing the mode weights to plot the evolution of the weights over a time-history.

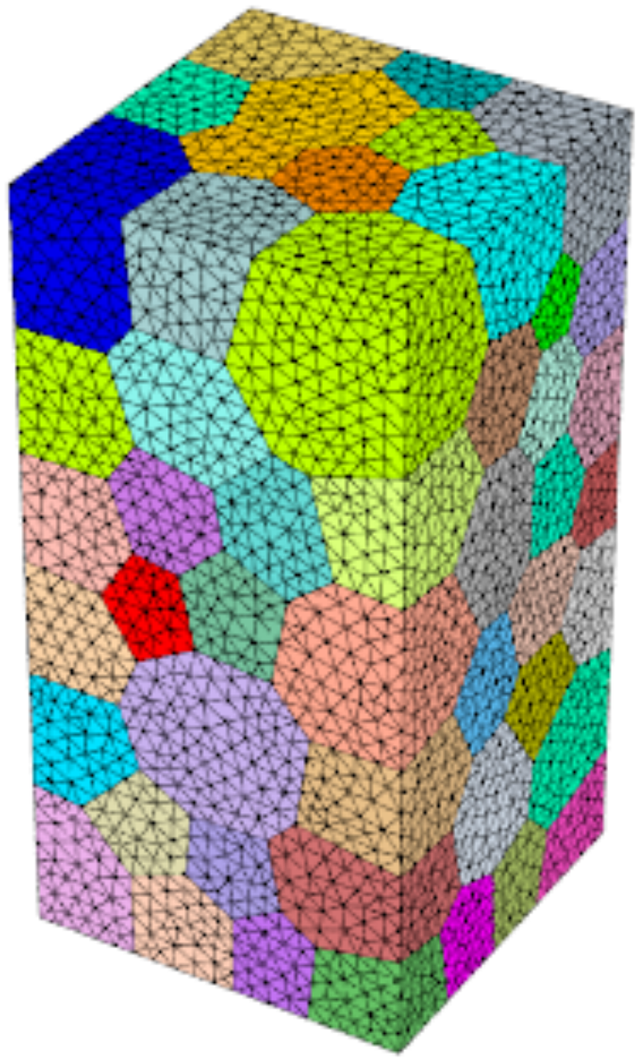
*Note: to use MechMonics with experimental stress data, see MechRecon.

# Example — Polycrystalline Tensile Sample

## Neper-Generated Virtual Polycrystal

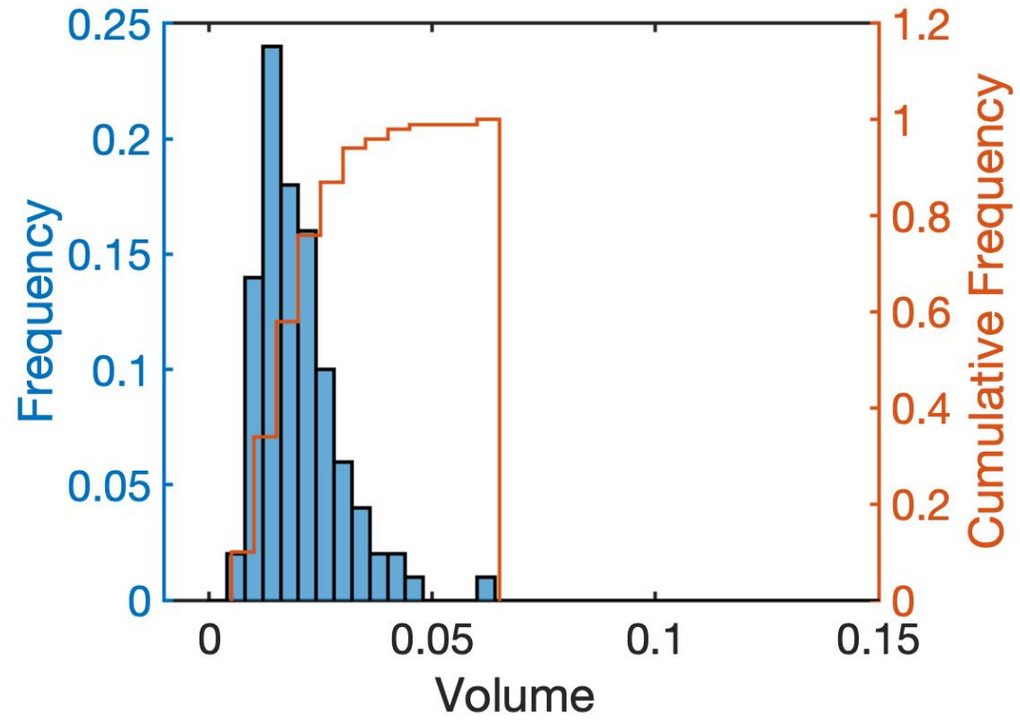
Sample attributes:

- 100 grains
- ~100,000 10-node tet elements
- ~180,000 nodal points
- (1:1:2) lengths for (x:y:z)
- Assigned S/X properties of FCC Al6XN (super-austenistic stainless steel)

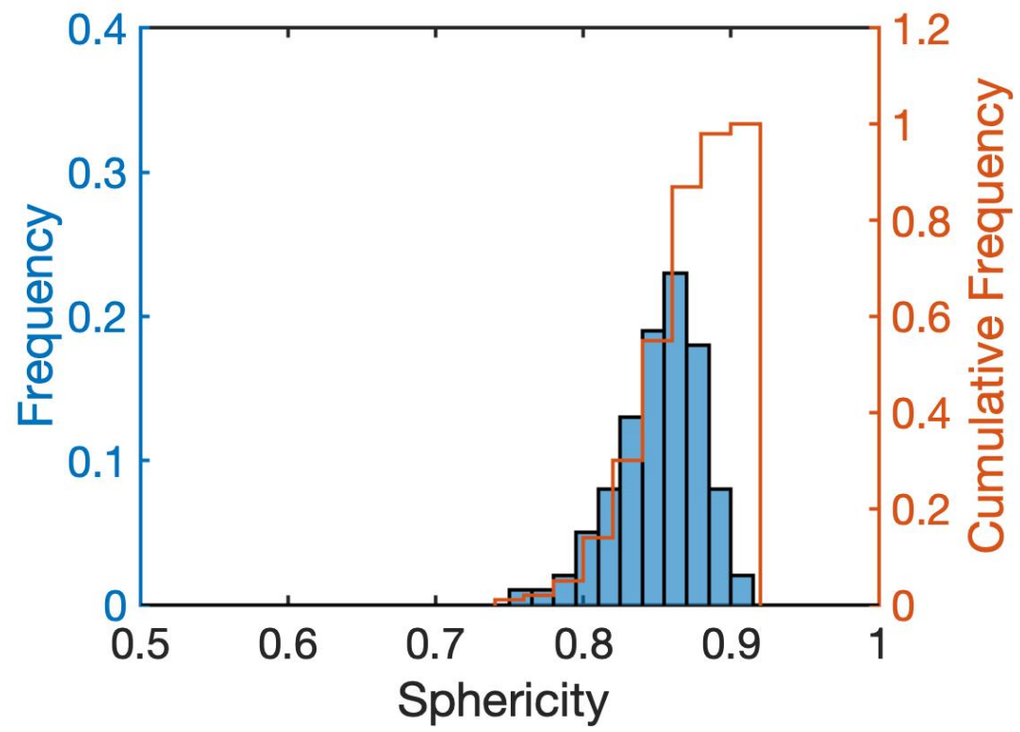


Grain statistics:

Grain size:



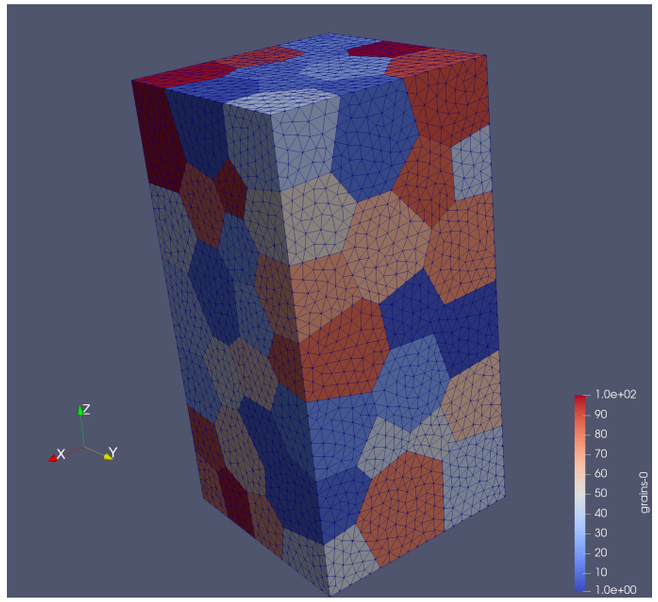

Grain sphericity:



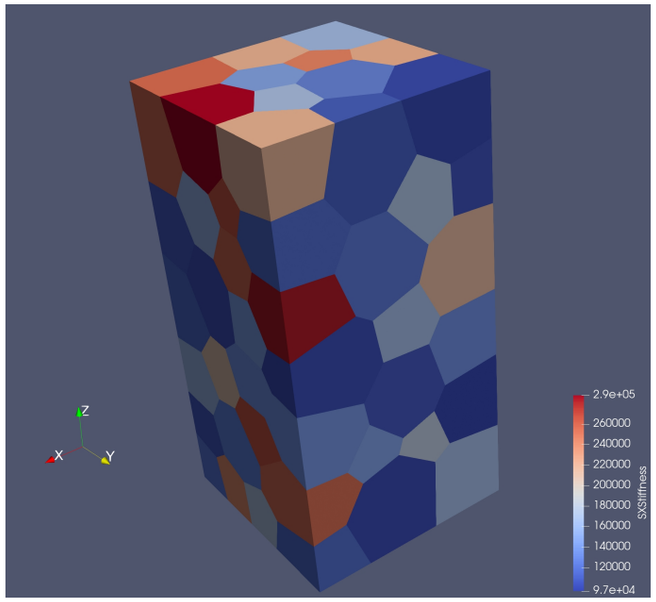
# Example — Polycrystalline Tensile Sample

## Mechanical Metrics Generated by MechMet

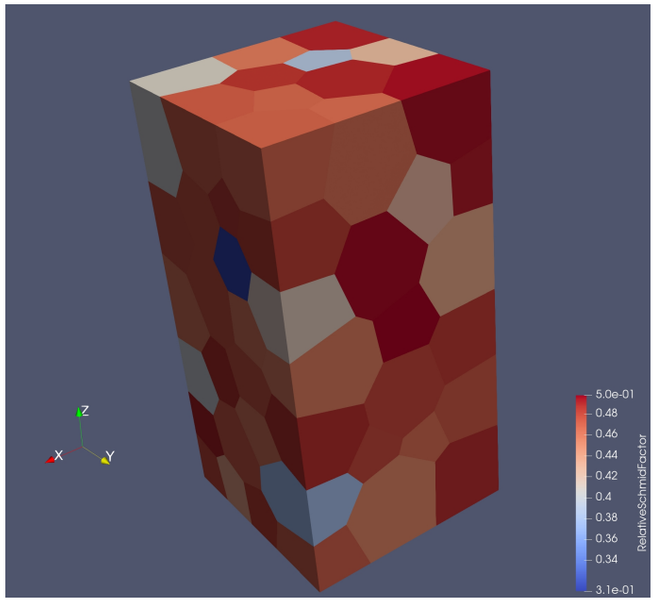
Grains



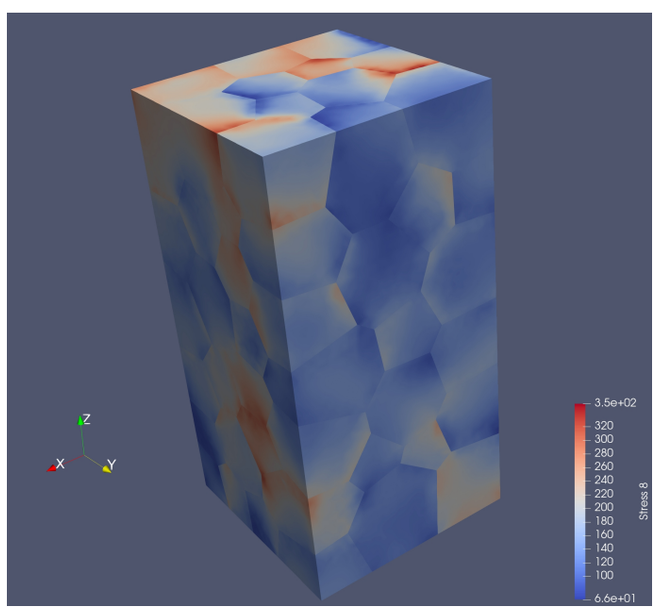

Directional elastic modulus



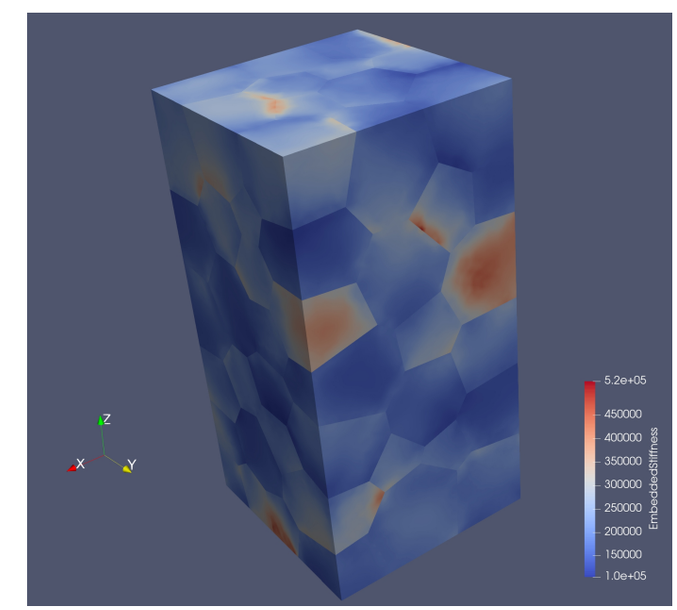

Schmid factor



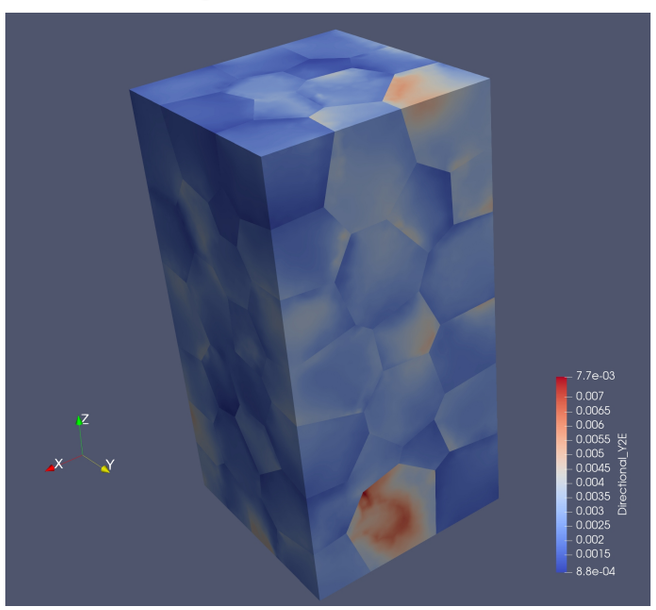

Axial stress at 0.1% extension in z-direction




Embedded (apparent) directional elastic modulus



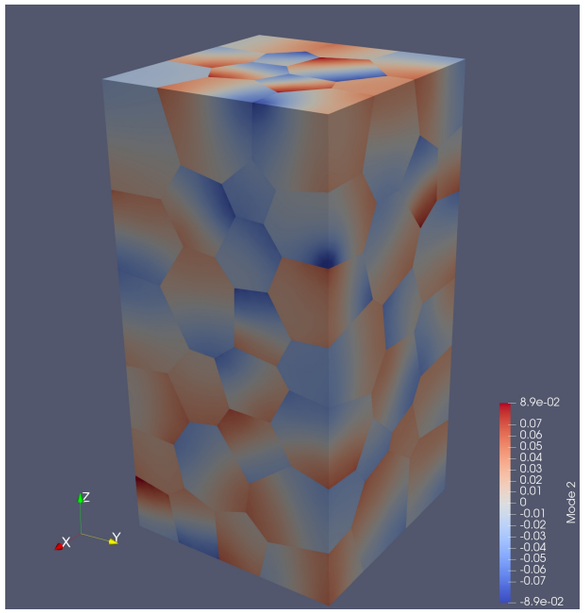

Directional strength-to-stiffness



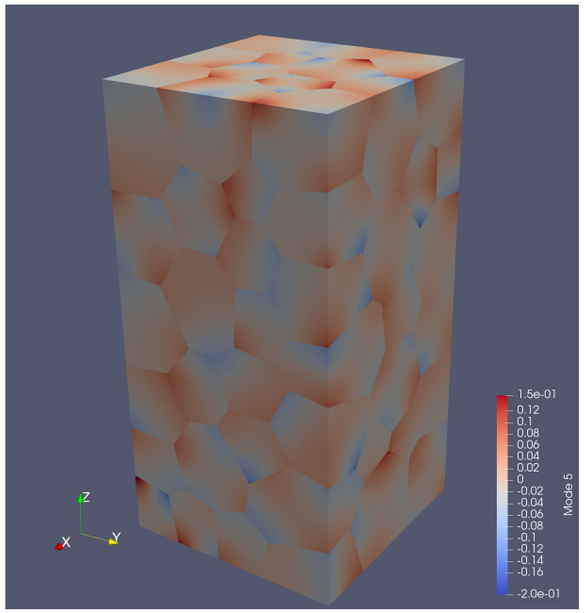
# Example — Polycrystalline Tensile Sample

## Three Harmonic Modes Generated by MechMonics

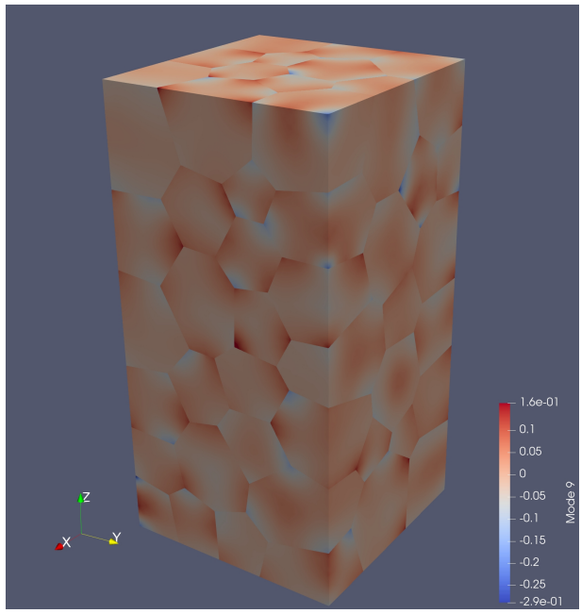
### Mode 2



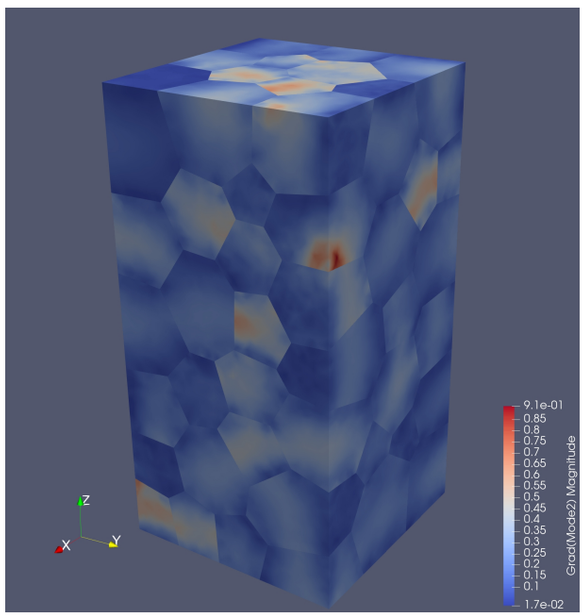

### Mode 5



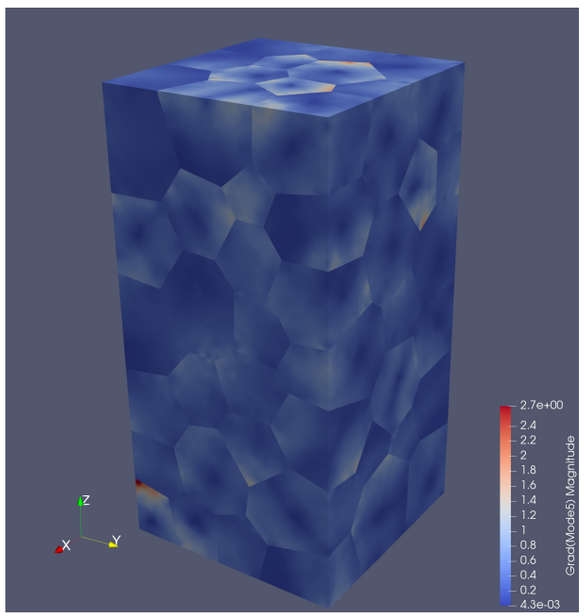

### Mode 9



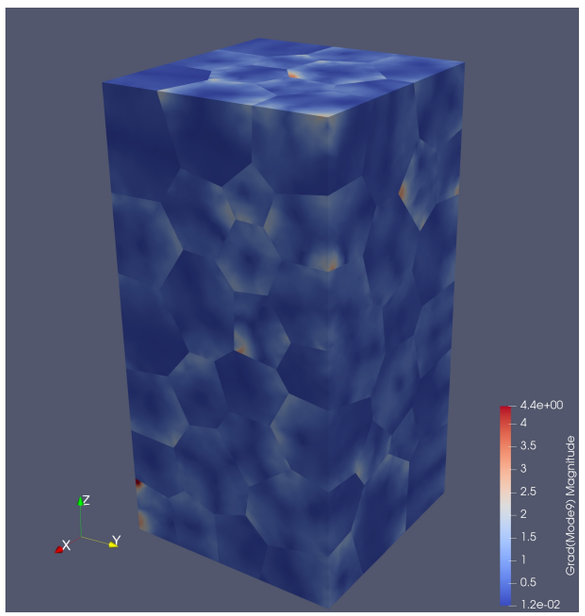

### mag(Grad(Mode 2))




### mag(Grad(Mode 5))



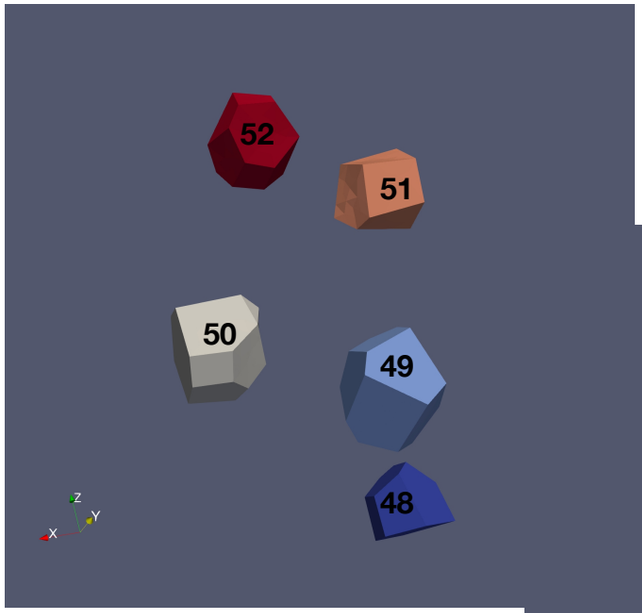

### mag(Grad(Mode 9))



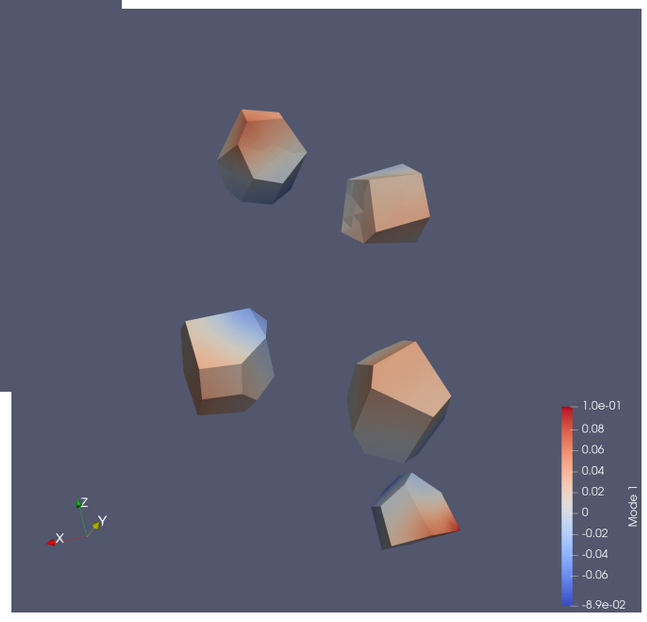

Mode 1 is a constant for each grain.

# Example — Polycrystalline Tensile Sample

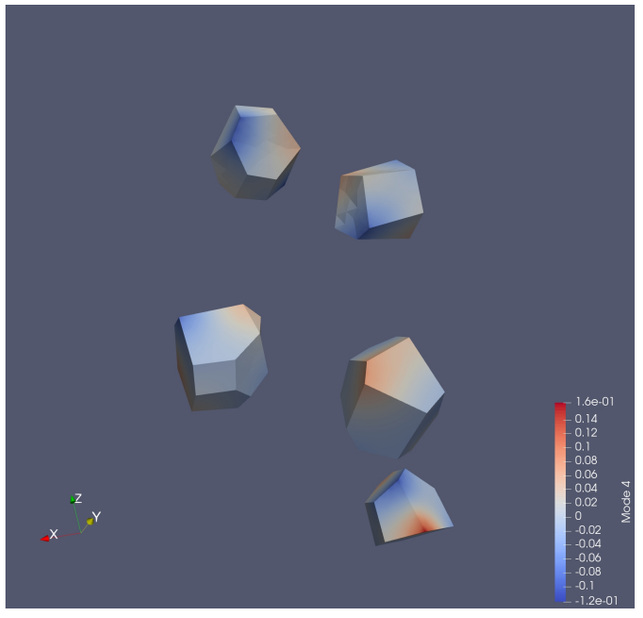
## Harmonic Modes in 5 Focus Grains (Grains 48-52)



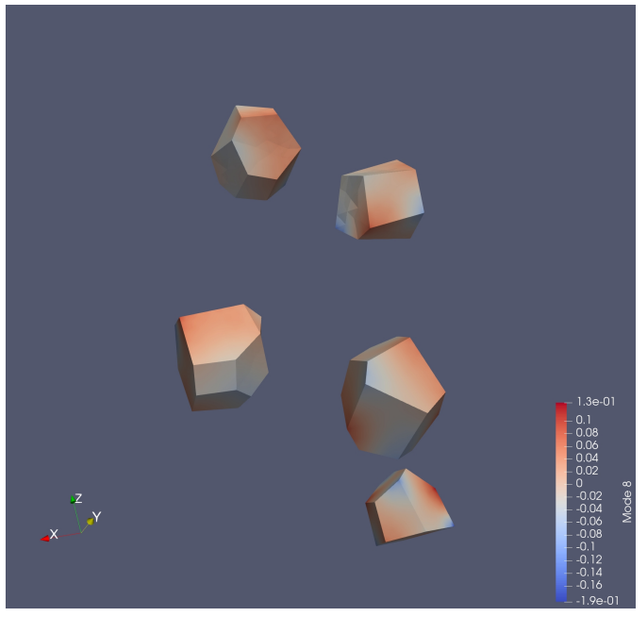

Mode 2



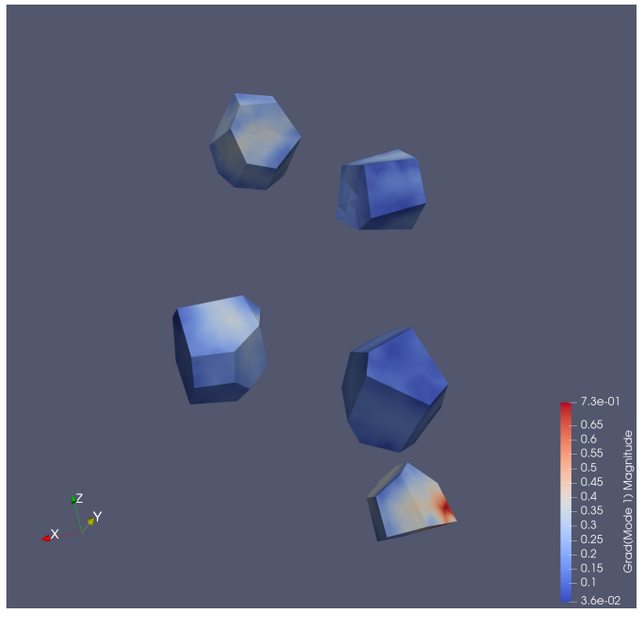

Mode 5



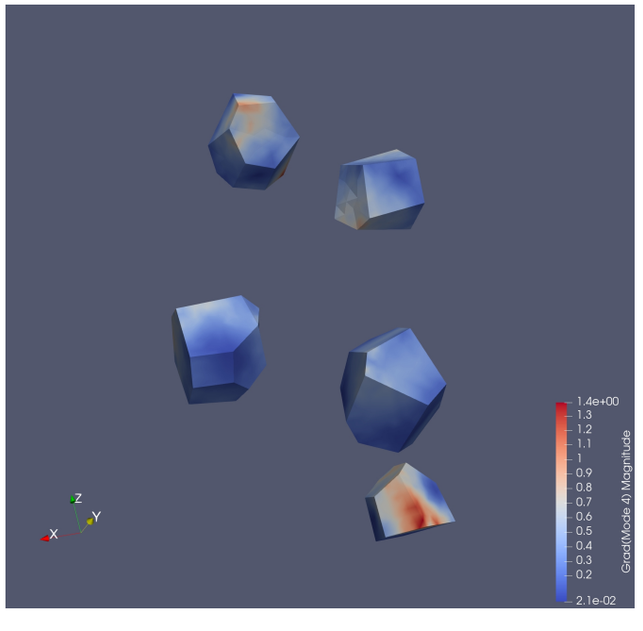

Mode 9



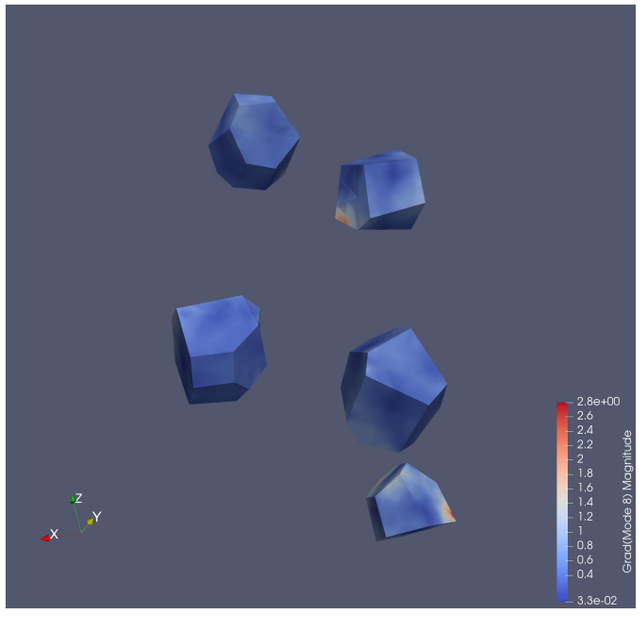

mag(Grad(Mode 2))




mag(Grad(Mode 5))



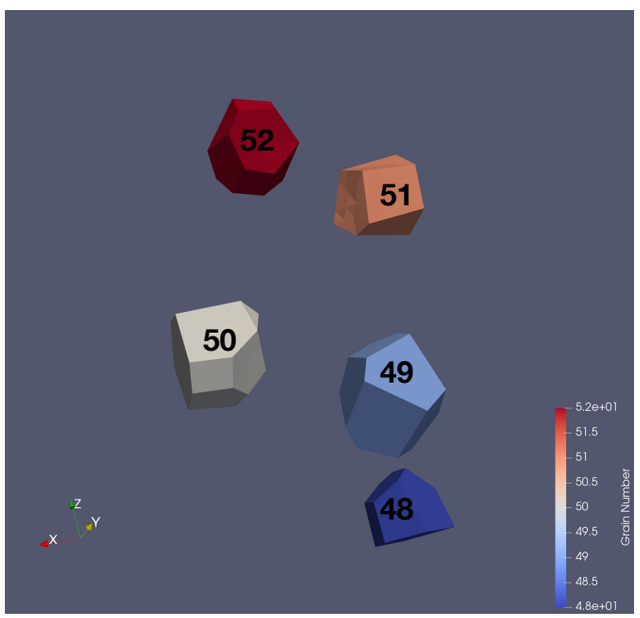

mag(Grad(Mode 9))



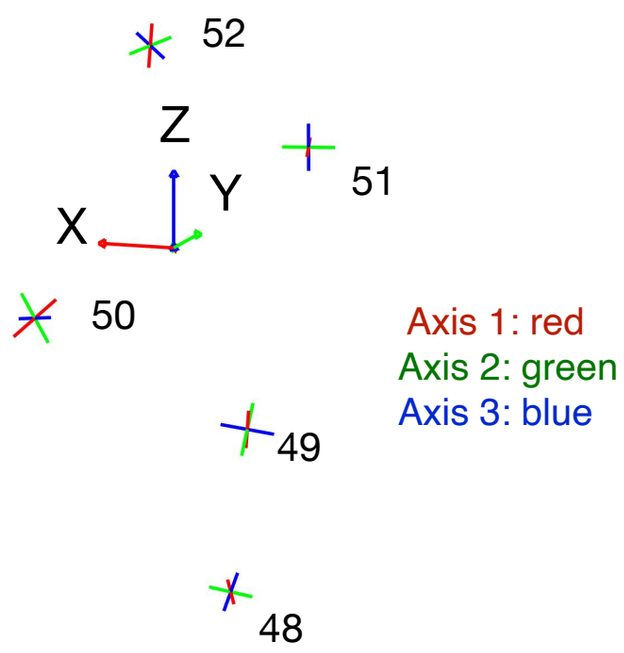
# Example — Polycrystalline Tensile Sample

## Grain Axes for the 5 Focus Grains (Grains 48-52)

Grains

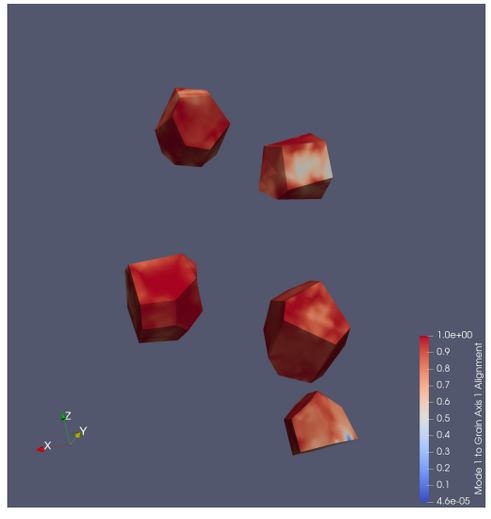



Grain Axes

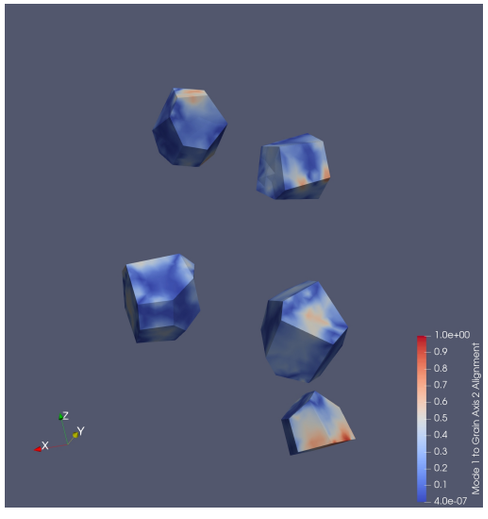



## Alignment of the Mode 1 Gradient Vector with the Grain Axes

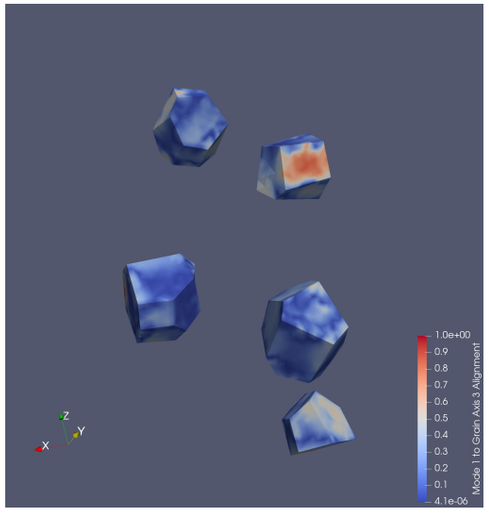
Axis 1




Axis 2




Axis 3

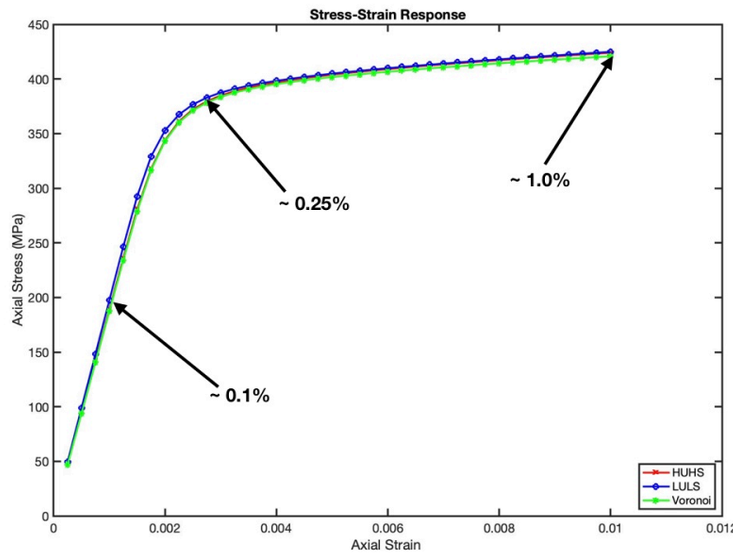


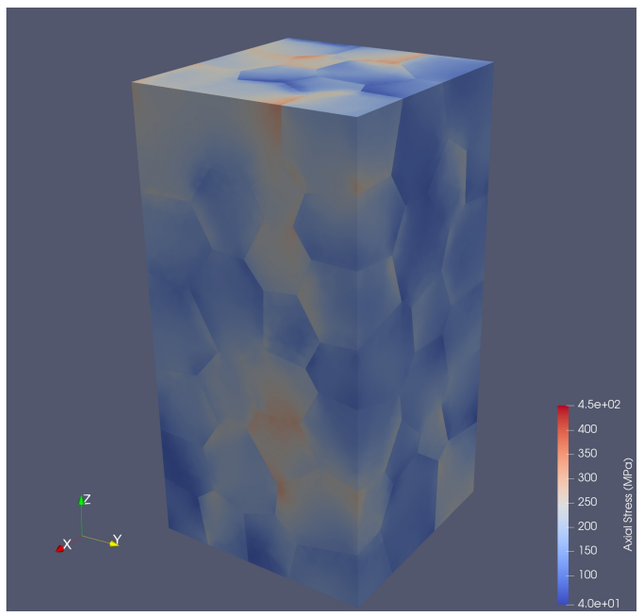
# Example — Polycrystalline Tensile Sample

## Field Data — Axial Stress Distributions Generated by FEpX

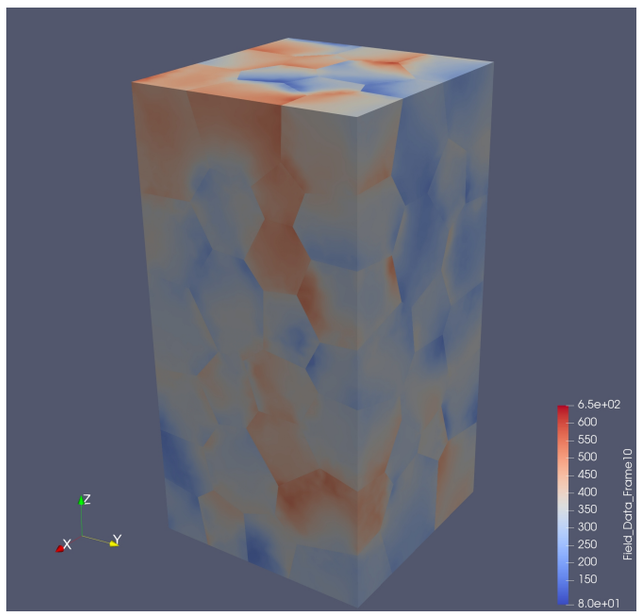
100 grain tensile sample
Mechanical properties for single-phase stainless alloy (fcc)
Sample extended to 1% nominal axial strain in increments of 0.025% axial strain.
Stress smoothing done grain-by-grain using quad point values

Stress-strain curve
(This sample is designated as HUHS on the plot legend)

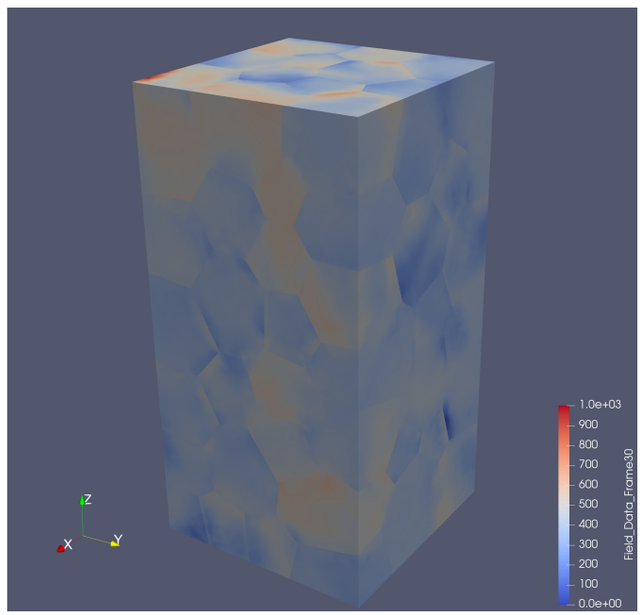



$\sigma_{zz}$ @ $\epsilon = 0.1\,\%$



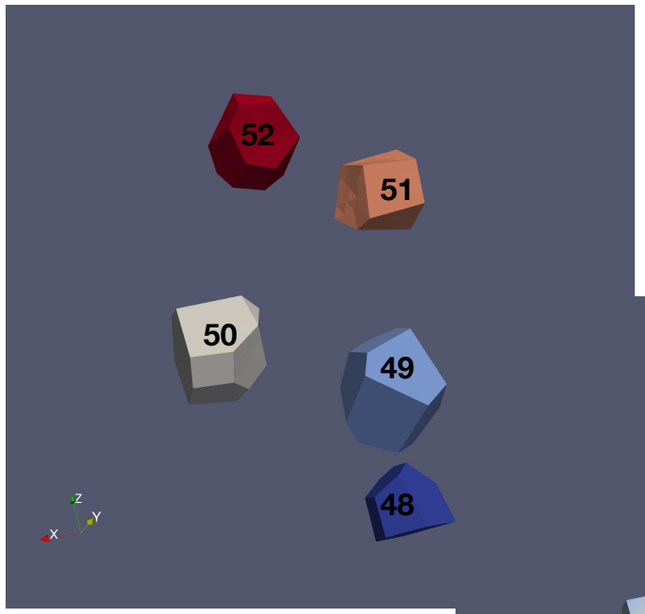

$\sigma_{zz}$ @ $\epsilon = 0.25\,\%$




$\sigma_{zz}$ @ $\epsilon = 1.0\,\%$

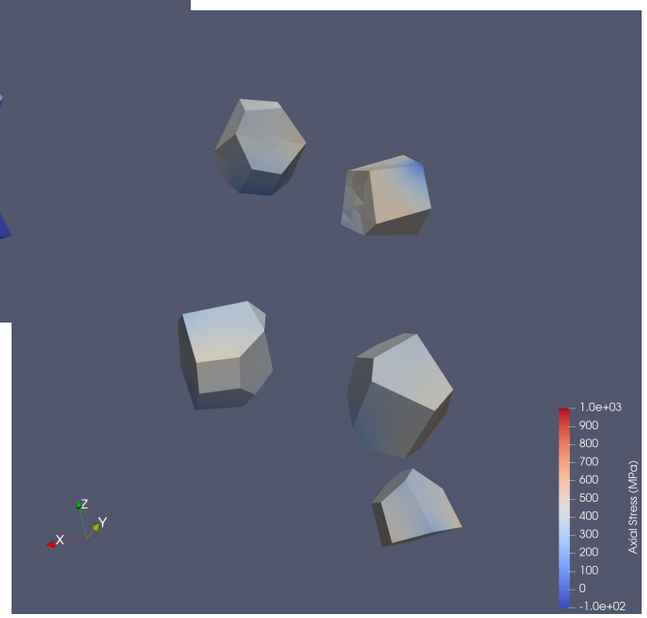
# Example — Polycrystalline Tensile Sample

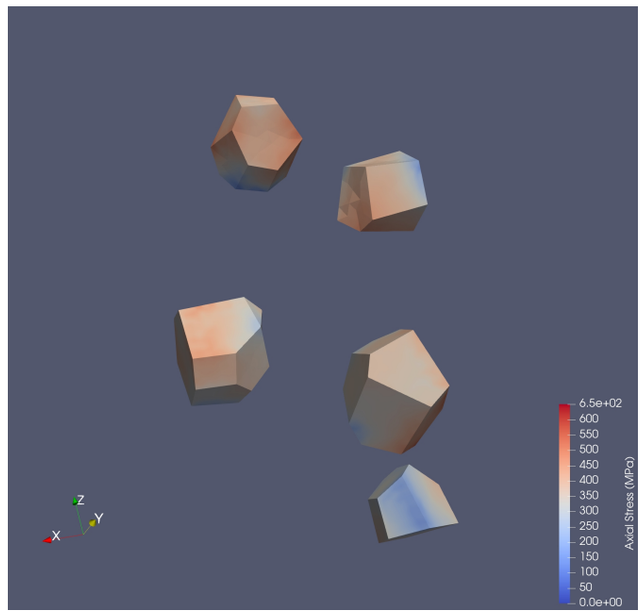
## Field Comparison of FEpX and Harmonic Expansion Distributions in 5 Focus Grains

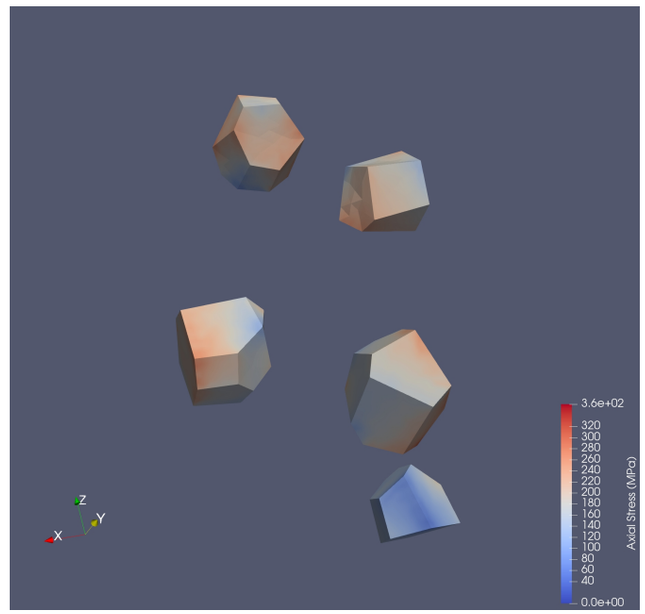



### Axial Stress Distributions

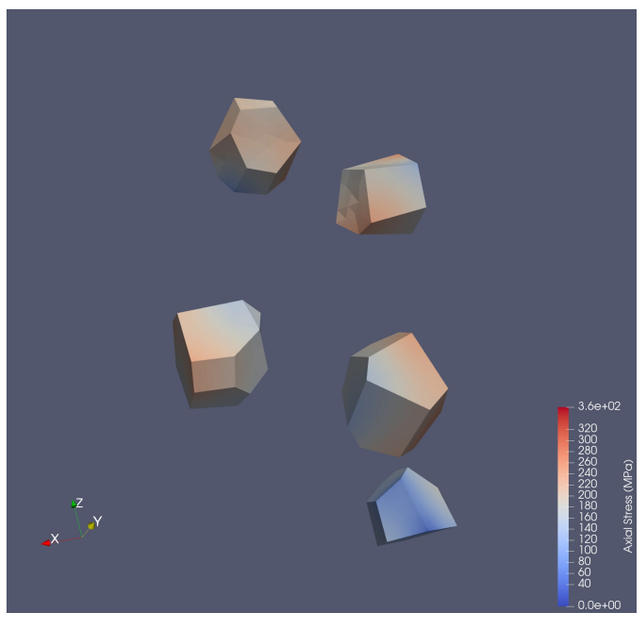
**FEpX (field data)**

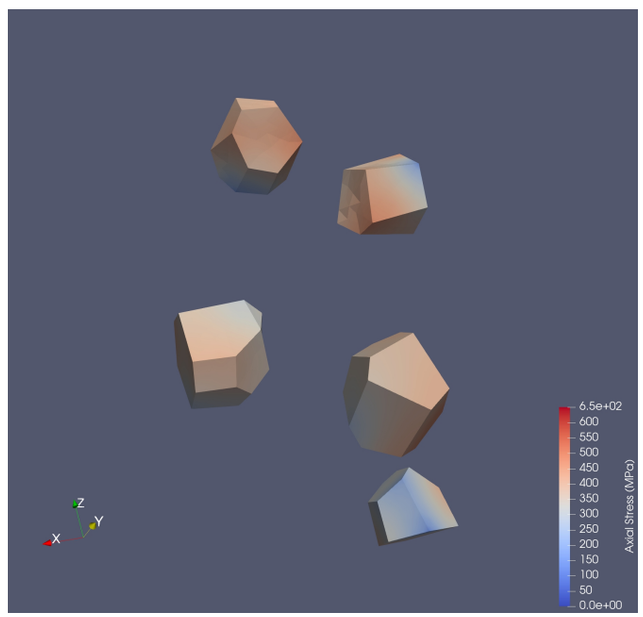


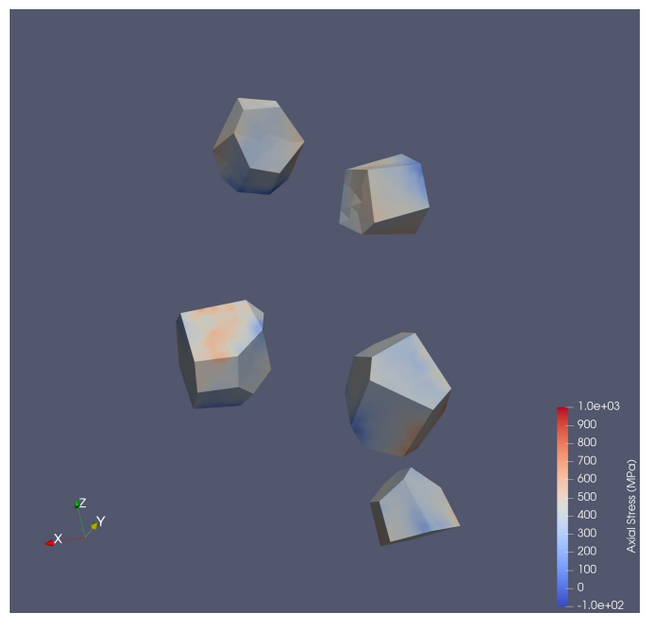




**Expansion (10 modes)**



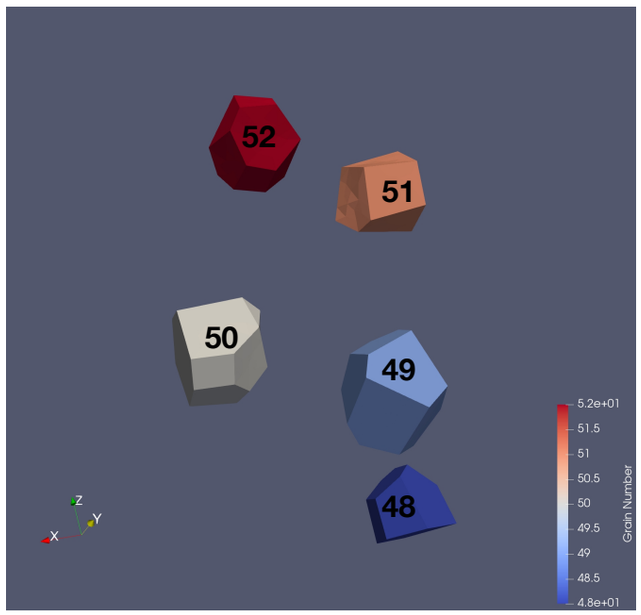



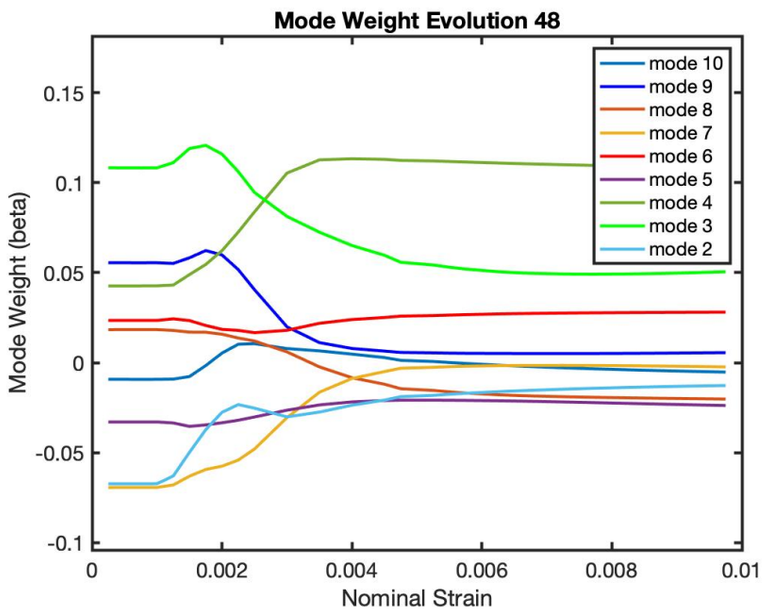

$\sigma_{zz}$ @ $\epsilon = 0.1\,\%$ $\sigma_{zz}$ @ $\epsilon = 0.25\,\%$ $\sigma_{zz}$ @ $\epsilon = 1.0\,\%$

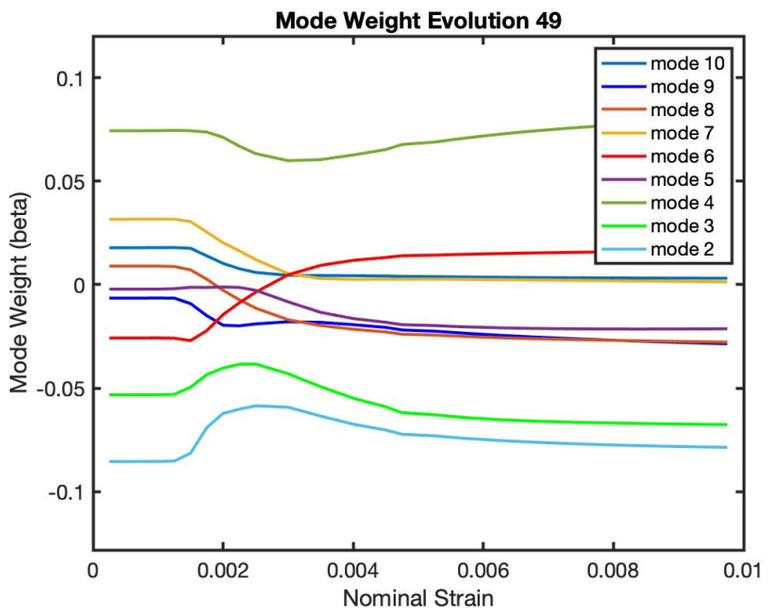
# Example — Polycrystalline Tensile Sample

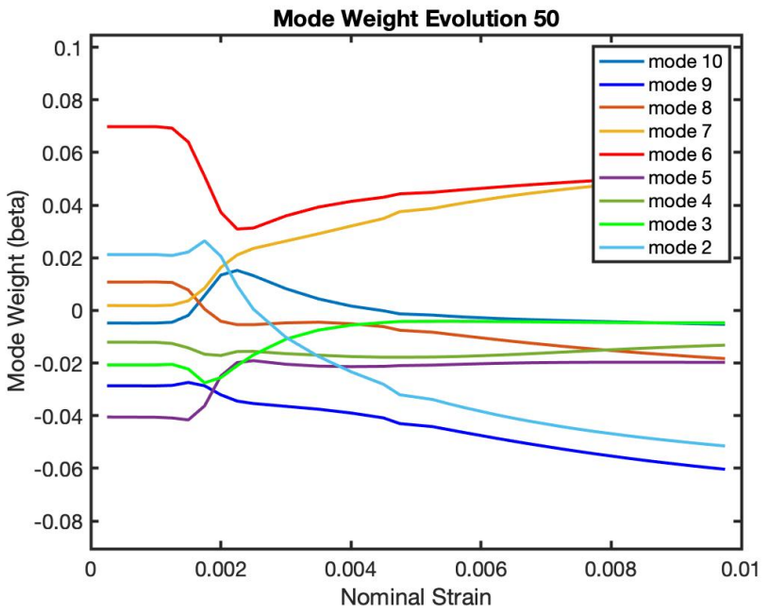
## Axial Stress Distributions

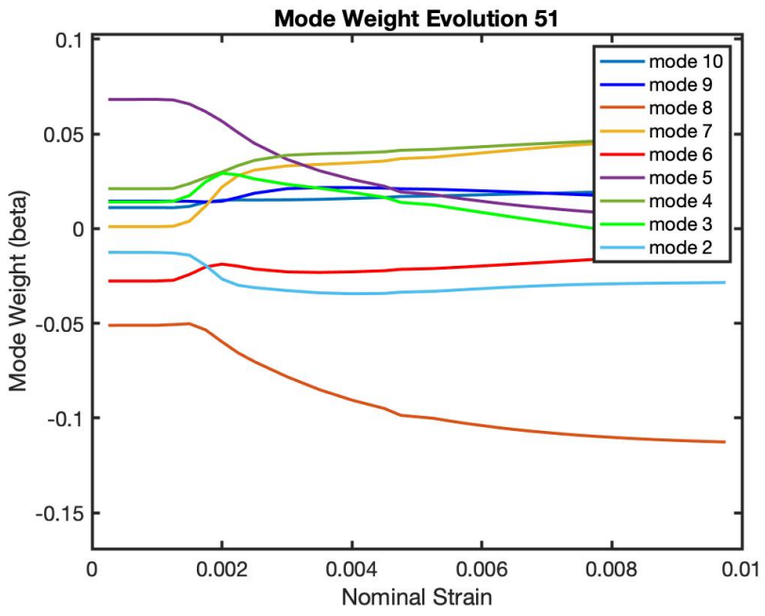
## Evolution of Harmonic Expansion Weights in 5 Focus Grains

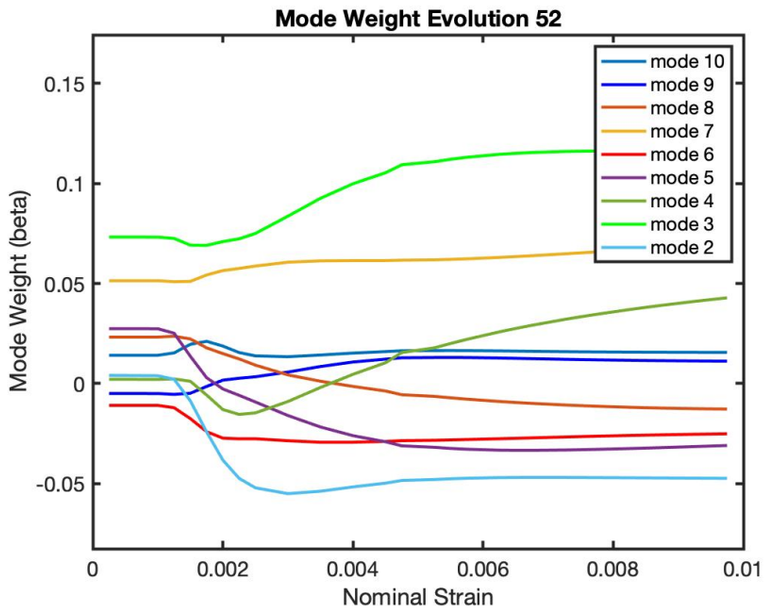



Weights are normalized by weight of Mode 1 (constant mode).